CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
First Semester of B. Pharm. Examination
University Theory Examination December 2015
PH114 Pharmaceutical Chemistry – I

Date: 14.12.2015, Monday **Time: 10.00 a.m. to 01.00 p.m.** **Maximum Marks: 80**

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat sketches wherever necessary.**

SECTION - I

Q 1 **Attempt all questions. Each question is of one mark.** **20**

1. **Decrease in secretion of acid in stomach is called as**

   [A] Hyperchlorhydria

   [B] Hypochlorhydria

   [C] Achlorhydria

   [D] Acidity

2. **Stannous fluoride is used as______________ in dental products.**

   [A] Anticaries agent

   [B] Dentifrice

   [C] Polishing Agent

   [D] Desensiting Agent

3. **Assay procedure of Zinc oxide is based on the principle of**

   [A] Acid-base titration

   [B] Redox titration

   [C] Complexometric titration

   [D] Gravimetric method

4. **____________ is used as titrant for the assay of potassium permanganate.**

   [A] 0.1 N hydrochloric acid

   [B] 0.1 N iodine

   [C] 0.1 N sodium thiosulphate

   [D] 0.1 N oxalic acid

5. Lugol's solution is composed of

   [A] Iodine

   [B] Iodine and Potassium iodate

[C] Iodine and Potassium iodide

[D] Iodine and Alcohol

6. Magnessium carbonate is assayed by

[A] Acid-base titration

[B] Redox titration

[C] Complexometric titration

[D] Precipitation titration

7. Limit tests are

[A] Quantitative or semi-quantitative tests

[B] Quantitative or qualitative tests

[C] Qualitative or semi-quantitative tests

[D] None of above

8. An alpha particle is

[A] An electron

[B] One neutron and one proton

[C] Two protons and two neutrons

[D] An X-ray emission

9. Which type of radioactive decay results in an increase in atomic number?

[A] positron emission

[B] alpha emission

[C] gamma emission

[D] beta emission

10. A photochemical reaction takes place by absorption of

[A] visible and UV radiation

[B] IR radiation

[C] heat energy

[D] none of these

11. "It is only the absorbed light radiations that are effective in producing a chemical reaction." This is the statement of

[A] Lambert law

[B] Lambert- Beer law

[C] Grothus- Draper law

[D] Stark- Einstein law

12. At constant temperature, the pressure of the gas is reduced to one third, the

[A] volumereduces to one third

[B] increases by three times

[C] remains the same

[D] can't be predicted

13. The unit in which surface tension is measured is

[A] dyne cm

[B] dyne cm^{-1}

[C] $dyne^{-1}$ cm

[D] $dyne^{-1}$ cm^{-1}

14. In Frenkel defect,

[A] some of the lattice sites are vacant

[B] an ion occupies interstitial position

[C] some of the cations are replaced by foreign ions

[D] none of the above

15. Physical adsorption occurs rapidly at ______ temperature.

[A] low

[B] high

[C] absolute zero

[D] room temperature

16. In chromatographic analysis, the principle used is

[A] absorption

[B] adsorption

[C] distribution

[D] evaporation

17. Which is not true for thermodynamics?

[A] it ignores the internal structure of atoms and molecules

[B] it involves the matter in bulk

[C] it is considered only with the initial and final states of the system

[D] it is not applicable to macroscopic systems

18. ______________ is not a manufacturing hazard.

[A] Inadequate storage condition

[B] Mislabelling

[C] Cross-contamination

[D] Microbial Contamination

19. Limit test for Iron is based on comparision of

[A] Opalescence

[B] Turbidity

[C] Violet Colour

[D] None of above

20. Which of the following is used as diluent in Limit test for sulphate?

[A] Tap water

[B] Mineral water

[C] Softened water

[D] Distilled water

SECTION – II

Q 2 Attempt any FOUR of the following

A	Enlist sources of impurities in Pharmaceutical Products. Discuss ANY TWO sources of impurities with relevant examples.	05
B	Write about principle, chemical reaction and uses of reagents for Limit Test of Chloride.	05
C	What is Pharmacopoeia? Describe its content.	05
D	What are Antimicrobial agents? Classify them based on mechanisms of action. Write ideal characteristics for Antimicrobial agents.	05
E	Write about general properties and assay of Boric acid.	05
F	Introduce the following class of drugs: (1) Protectives (2) Saline Cathartics (3) Dentifrices (4) Expectorants (5) Adsorbents	05

SECTION – III

Q 3 Attempt any FOUR of the following

A	What are Antacid? Give its detailed classification. Write a brief note on Aluminium hydroxide gel.	05
B	Write a short note on Antidotes.	05
C	What is the physiological role of Iron and Copper in human body. Enlist official compounds of Iron and Copper.	05
D	Draw and explain Jablonski Diagram.	05
E	Draw the schematic diagram of the instrument used for measurement of intensity of transmitted light and explain various detectors used for the same.	05
F	State and explain the gas laws.	05

Q 4 Attempt any FOUR of the following

A	What do you mean by colligative property? Explain any one of them in detail.	05
B	Add a detailed note on crystal defects.	05
C	Write a note on adsorption isotherm.	05
D	Explain various types of nuclear reaction with suitable examples.	05
E	State various laws of thermodynamics and differentiate reversible and irreversible processes.	05
F	Enumerate various methods of measurement of radioactivity and explain any two of them with suitable diagrams	05
